MASTER OF BUSINESS ADMINISTRATION

SEMESTER 3

DADS304

VISUALIZATION

Unit 3

Introduction to R SHINY

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1. INTRODUCTION

In the developing field of **visual analytics**, interactive visual interfaces are used to facilitate analytical reasoning. The fundamental concept is the combination of exceptional human abilities for visual information exploration with enormous computing power to create a potent environment for knowledge discovery.

Other visual analytics programmes like Flourish, Infogram, D3, and many others have the issue of either being expensive to use or not being sophisticated enough to be used with more complex statistical analysis programmes like dynamical modeling ones.

R Shiny is a free and open-source component of the R programming language, and as such, it is integrated with R's enormous array of statistical, numerical, and computational capabilities.

INSPIRED BY LIFE

2. VISUAL ANALYTICS

The use of complex methods and tools to evaluate data using graphical representations of the information is known as visual analytics. Users can see patterns and gain useful insights by viewing the data as graphs, charts, and maps. Organizations can improve their data-driven decisions thanks to these insights.

Benefits:

Share findings and monitor progress:Organize and share key performance indicators across an organisation by using interactive reports and dashboards.

Take action more quickly: When working with data sets in a visual format, users can comprehend data insights much more quickly.

Easier Data Exploration:Without requiring assistance from IT, self-service analytics solutions that let users engage with data in a visual context enable them to find hidden links and patterns.

Encourages Data Literacy: Data analytics becomes more accessible by making data easy to use and comprehend, involving more individuals inside a company.

Difference between data analytics and data visualization

The term "data visualisation" typically refers to the graphical representation of data, or representing data in bubble charts, heat maps, and other visuals to aid in understanding patterns, relationships, trends, and other important insights in datasets. The term "visual analytics" describes the use of an analytics tool to carry out in-depth analysis of large, complicated datasets while enabling users to interact and explore dynamic visuals.

Visual Analytics Best Practices

- Define goals
- Integrate and manage the data
- Simplify visualizations
- Get Inspired

Examples:

- Marketing:By enabling the marketer in this example to see and comprehend each stage of the customer life cycle, visual analytics helps them increase ROI.
- Supply Chain:By displaying KPIs and enabling interactive exploration, big data visual analytics can assist supply chain managers in quickly discovering relationships across complicated, divergent data sources.
- Sales:The clear, organised presentation of sales data helps sales managers to boost revenue, enhance forecasting, and spot important patterns.
- Finance:A loan manager at a consumer bank can investigate how various geographic areas, products, and loan officers fare over time and determine which factors have the most influence on revenue and profits.
- IT :Data analytics and visualisation can be used by IT administrators to better predict future technology requirements and spot underutilized systems and applications.

Self-Assessment Questions - 1

1. Users can see patterns and gain useful insights by viewing the data as _____, _____ , and _____.
2. The _____ describes the use of an analytics tool to carry out in-depth analysis of large, complicated datasets.

3. INTRODUCTION TO R-SHINY

Imagine being able to create a web application using your #datascience analysis. The R Shiny programme allows you to do exactly that.

Using the wonderful R Shiny framework, you can quickly turn your data research into a web app. Create incredible apps that your company can utilize in a matter of hours, not weeks or months.

An R package called Shiny makes it possible to create interactive web applications that can run R code in the background. With Shiny, you can create dashboards, embed interactive charts in R Markdown papers, and host standalone applications on a website. Additionally, you can add HTML widgets, JavaScript actions, and CSS themes to your Shiny applications.

The core functionality of the Shiny web framework is the ability to gather input values from a web page, make those inputs readily available to the application in R, and have the output values from the R code posted to the web page. A Shiny application needs a user interface and a server function to do calculations in its most basic version. A server script and a user-interface definition make up Shiny applications' two parts.

TOOLS

Installing the following tools is necessary:

- R: Located here: https://cran.r-project.org/bin/windows/base/.
- R Studio We strongly advise you to download RStudio 1.4.
- Install.packages("shiny"): latest stable release
- Install "tidyverse" into your packages.
- A cutting-edge web browser It's best to use Google Chrome.

Self-Assessment Questions - 2

3. _____ can be done using R Shiny for data science.
4. There are 2 phases of R-Shiny Applications (True/False).

4. WHAT IS R SHINY?

An R tool called Shiny makes it simple to create dynamic web applications directly from R. You can create dashboards, embed standalone apps in R Markdown papers, or host them on a website.

Installation:

Installing the package is a prerequisite for using it in R programming. Install.packages("packagename") is the command that can be used to do this task. Type this to install the entire Shiny package:

install.packages("shiny")

FEATURES

- Create straightforward web applications without JavaScript
- Similar to how spreadsheets are automatically "active." Changes to the input will result in an automated update.
- Pretty default UI style based on Twitter Bootstrap; works in any R environment.
- Display plots, tables, and printed output from R objects using pre-made and editable output widgets.

Self-Assessment Questions - 3

5. We install Shiny by using the command _____.
6. R-Shiny helps to create straightforward web apps without _____.

5. CREATING BASIC R SHINY APP

Shiny programmes have a .R extension at the end, much like R files do. There are three parts to the app structure, which are as follows:

1. a user interface object (ui)
2. a server function
3. A function call to shinyApp

Radio buttons, panels, and selection boxes are all managed by the user interface (ui) object, which is used to manage the app's overall design and layout.

UI OBJECT

The fluidPage() method is used to construct the app's layout for the default app's user interface. The fluidPage layout will adapt to changes in browser size automatically. The fluidPage() uses a sidebarLayout() to divide the page into two sections: the mainPanel(), which houses the histogram output in the example app, and the sidebarPanel(), which houses the app's input (the slider). This is a reasonably standard layout for a Shiny programme; take note that nothing prevents us from adding additional output before or after the sidebarLayout(), such as the titlePanel(). Each page component, with fluidPage() at the top, is an argument to a function.

There are more advanced layout options that are also more flexible, like navBarPage(), to create a page with a navigation bar. A user interface can also be created entirely from scratch using HTML, CSS, etc.

The web application's code is located in the server function. A render...() function corresponds to each...Output() function in the user interface. The R code to create the object we wish to render is provided as the first argument to the render...() function. For instance, the code to create a histogram is located in the renderPlot() function of the example app. Since this must be a single expression, it is usually found in s.

```
ui <- fluidPage(

   # Application title
   titlePanel("Old Faithful Geyser Data"),
   p("Here is some text"),
   # Sidebar with a slider input for number of bins
```

Fig.1 Implementation of titlePanel()

The value of the slider as defined in the user interface will be contained in input$bins (which was called bins).

```
sliderInput("bins",
            "Number of bins:",
            min = 1,
            max = 50,
            value = 30)
```

Fig.2 Implementation of sliderInput()

A SERVER FUNCTION

The web application's code is located in the server function. A render...() function corresponds to each...Output() function in the user interface. The R code to create the object we wish to draw is provided as the first parameter to the render...() function. For instance, the code to create a histogram is located in the renderPlot() method of the sample app. Since this must be a single expression, it is usually seen in s.

```
   output$distPlot <- renderPlot({
      # generate bins based on input$bins from ui.R
      x    <- faithful[, 2]
      bins <- seq(min(x), max(x), length.out = input$bins + 1)

      # draw the histogram with the specified number of bins
      hist(x, breaks = bins, col = 'darkgray', border = 'white')
   })
```

Fig.3 code for plotting

REACTIVITY-Responding events by Shiny

Reactive programming is the method used by Shiny. This implies that everything that depends on something will immediately be updated when that item changes (such the slider being adjusted). The distPlot graph is the only component of the sample app that is dependent on the slider; as a result, if we change the slider, this will be redrew.

The graph is also influenced by several other, less obvious aspects of the programme, such the window size. The graph will be re-drew if the window is resized since Shiny is aware that the graph depends on that aspect of the app.

SAMPLE CODE OF A GAME:

UI:

First page can be done using simple HTML features like:

```
intro_panel <- tabPanel(
  "[Tab Title]",

  titlePanel("[Page Title]"),

  img(src = "[img source]"),

  p("[Summary text for page]"),

  p(a(href = "[url]", "[Link text]"))
)
```

Fig.4 sample code for image insertion

The second panel, which will house all of our visualisation work, will then be defined. We must specify our main content and sidebar content before combining them in our second tabPanel ().

We'll add a choose widget to our sidebar so the user may choose the Y variable for our plot. This choose widget will be given the name "y var," and we'll use it later to modify the plot on

our server. R file. We use the term "plot" in our primary material but define it afterwards on our server ,also a R file.

Call select values = colnames to set the select values variable for the selectInput choices (data).

```
sidebar_content <- sidebarPanel(
  selectInput(
    "y_var",
    label = "Y Variable",
    choices = select_values,
    selected = "Speed"
  )
)

main_content <- mainPanel(
  plotOutput("plot")
)
```

Fig.5 sample code

Design the second panel also using HTML features.

```
second_panel <- tabPanel(
  "[Tab Title]",

  titlePanel("[Page Title]"),

  sidebarLayout(
    sidebar_content, main_content
  )
)
```

Fig.6 panel using HTML

Finally we combine the pages in the navbar object.

```
ui <- navbarPage(
  "[Nav Bar Title]",
  intro_panel,
  second_panel
)
```

Fig.7 combined pages in navbar object

SERVER:

1. Load data files and libraries.
2. Gather data
3. Create graphs and charts using the server function.

The function that assigns values to the 'output', as shown below, will be created in the server.R file. It will accept input values specified by the UI.

To match the "plot" label we wrote in our UI main panel, we want to specify output$plot in our server function as follows: plotOutput("plot"). RenderPlot() will be called to construct the plot in order to initialise this.

```
server <- function(input, output) {
  # assign values to `output` here
}
```

```
output$plot <- renderPlot({

   ggplot(data=characters, aes_string(x='Character',
      y=input$y_var, fill="Class")) +
      geom_bar(stat="identity", width=0.8) +
      labs(x="Character", y=input$y_var) + coord_flip()

})
```

Fig.8 output for function

The main aspect is incorporating your supplied variables. We created the UI input variable y var to be utilised in choosing which variable appears in the y-axis (in a vertical bar plot, it will show as the x-axis). This variable, which you may refer to as input$y var, is used to organise, label, and show data in your plot.

The connections have been made, so you may view and use your visualisation now.

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The connections have been made, so you may view and use your visualisation now.

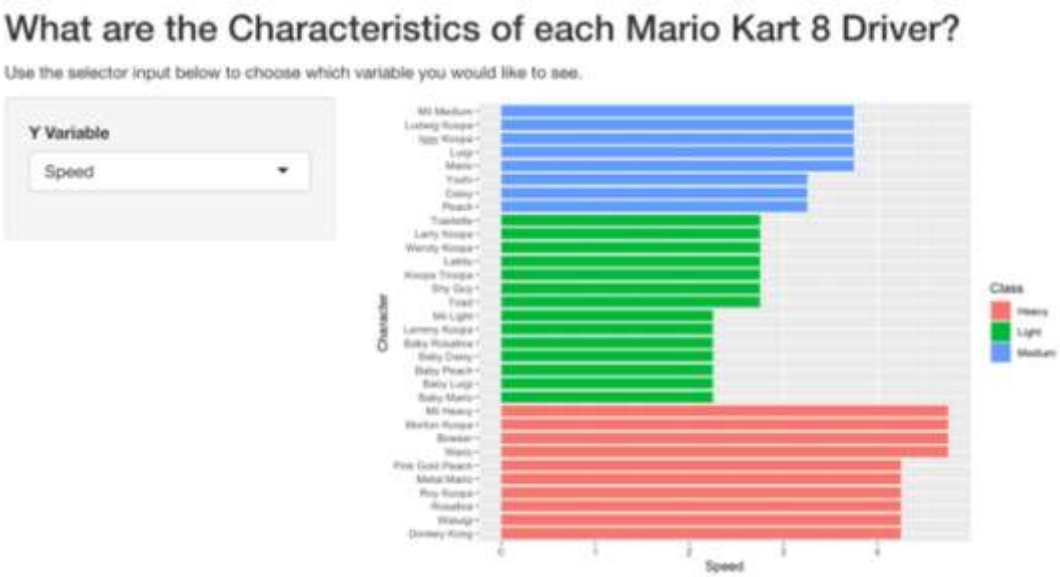

Fig.9 demonstration of mario kart 8 driver

Self-Assessment Questions - 4

7. The server phase in R-Shiny Load data files and libraries , gathers data and _____.
8. _____ features are used to design the UI interface.

6. SUMMARY

In the developing field of **visual analytics**, interactive visual interfaces are used to facilitate analytical reasoning. The fundamental concept is the combination of exceptional human abilities for visual information exploration with enormous computing power to create a potent environment for knowledge discovery.

An R package called Shiny makes it possible to create interactive web applications that can run R code in the background. With Shiny, you can create dashboards, embed interactive charts in R Markdown papers, and host standalone applications on a website. In addition , one can add HTML widgets, JavaScript actions, and CSS themes to your Shiny applications.

7.ACTIVITY

Take the gapminder data, examine it, then make an amusing app using r shiny to show the dataset for fun.

Dataset can be downloaded from kaggle.

8. GLOSSARY

*R-Shiny:*The open source R package that offers a beautiful and robust web framework for creating online apps.

Visual Analytics: The use of advanced tools and procedures to evaluate datasets using visual representations of the data is known as visual analytics.

9. STUDY NOTES & DID YOU KNOW

- Shiny is a web app tool, joining others like Streamlit and Dash.
- Currently in alpha, Shiny for Python will likely undergo significant changes over the next few months. However, we can already test it out and see how it performs.

10. CASE STUDY

Let's look at this case study in order to better comprehend R shine.

Here, we examine data from the Consumer Product Safety Commission's (CPSC) National Electronic Injury Surveillance System (NEISS).

```
library(shiny)
library(vroom)
library(tidyverse)
```

Fig.10 importing modules

```
dir.create("neiss")
#> Warning in dir.create("neiss"): 'neiss' already exists
download <- function(name) {
  url <- "https://github.com/hadley/mastering-shiny/raw/master/neiss/"
  download.file(paste0(url, name), paste0("neiss/", name), quiet = TRUE)
}
download("injuries.tsv.gz")
download("population.tsv")
download("products.tsv")
```

Fig.11 To load data

```
injuries <- vroom::vroom("neiss/injuries.tsv.gz")
injuries
#> # A tibble: 255,064 × 10
#>   trmt_date    age sex   race  body_part   diag    locat…¹ prod_…² weight
#>   <date>     <dbl> <chr> <chr> <chr>       <chr>   <chr>     <dbl>  <dbl>
#> 1 2017-01-01    71 male  white Upper Trunk Contus… Other …    1807   77.7
#> 2 2017-01-01    16 male  white Lower Arm   Burns,… Home        676   77.7
#> 3 2017-01-01    58 male  white Upper Trunk Contus… Home        649   77.7
#> 4 2017-01-01    21 male  white Lower Trunk Strain… Home       4076   77.7
#> 5 2017-01-01    54 male  white Head        Inter … Other …    1807   77.7
#> 6 2017-01-01    21 male  white Hand        Fractu… Home       1884   77.7
#> # … with 255,058 more rows, and abbreviated variable names ¹location,
#> #   ²prod_code, ³narrative
```

Fig.12 Sample data in the dataset

```
products <- vroom::vroom("neiss/products.tsv")
products
#> # A tibble: 38 × 2
#>   prod_code title
#>       <dbl> <chr>
#> 1       464 knives, not elsewhere classified
#> 2       474 tableware and accessories
#> 3       604 desks, chests, bureaus or buffets
#> 4       611 bathtubs or showers
#> 5       649 toilets
#> 6       676 rugs or carpets, not specified
#> # … with 32 more rows

population <- vroom::vroom("neiss/population.tsv")
population
#> # A tibble: 170 × 3
#>     age sex    population
#>   <dbl> <chr>       <dbl>
#> 1     0 female    1924145
#> 2     0 male      2015150
#> 3     1 female    1943534
#> 4     1 male      2031718
#> 5     2 female    1965150
#> 6     2 male      2056625
#> # … with 164 more rows
```

Fig.13 Pairing of two more data for context

```
selected <- injuries %>% filter(prod_code == 649)
nrow(selected)
#> [1] 2993
```

Fig.14 Exploration of the data

```
summary <- selected %>%
  count(age, sex, wt = weight)
summary
#> # A tibble: 208 × 3
#>     age sex         n
#>   <dbl> <chr>   <dbl>
#> 1     0 female   4.76
#> 2     0 male    14.3
#> 3     1 female 253.
#> 4     1 male   231.
#> 5     2 female 438.
#> 6     2 male   632.
#> # … with 202 more rows

summary %>%
  ggplot(aes(age, n, colour = sex)) +
  geom_line() +
  labs(y = "Estimated number of injuries")
```

Fig.15 Summary

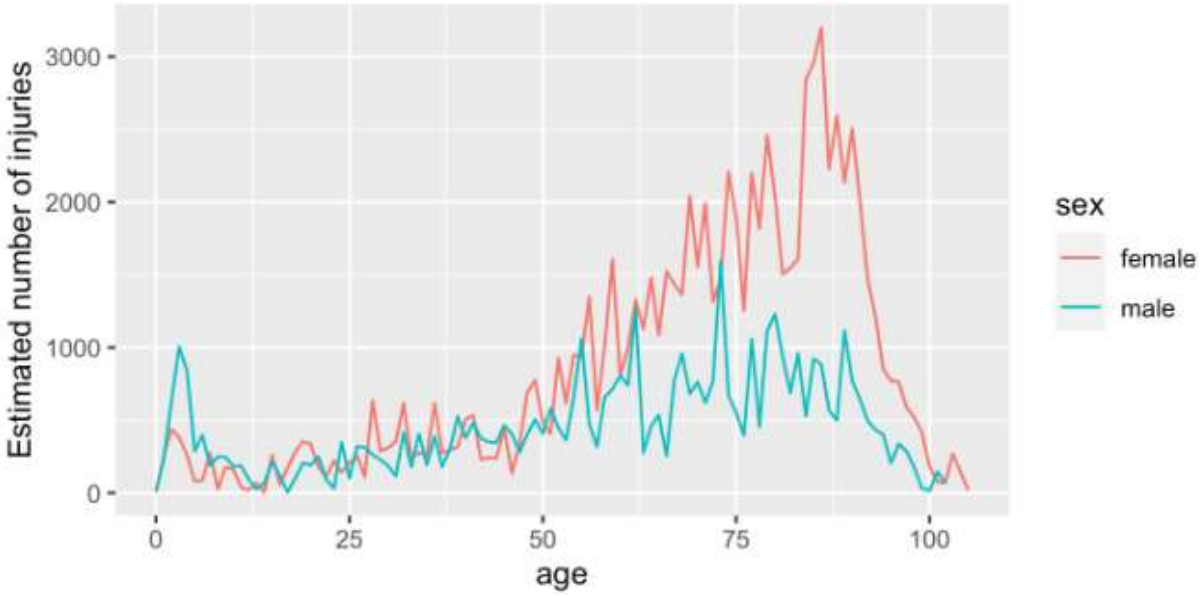

Fig.16 Estimated number of injuries according to age of both male and female

```
prod_codes <- setNames(products$prod_code, products$title)

ui <- fluidPage(
  fluidRow(
    column(6,
      selectInput("code", "Product", choices = prod_codes)
    )
  ),
  fluidRow(
    column(4, tableOutput("diag")),
    column(4, tableOutput("body_part")),
    column(4, tableOutput("location"))
  ),
  fluidRow(
    column(12, plotOutput("age_sex"))
  )
)
```

Fig.17 Prototype

Product

knives, not elsewhere classified

diag	n
Laceration	287925.65
Avulsion	9970.11
Puncture	5870.19
Other Or Not Stated	4852.37
Contusion Or Abrasion	1687.99
Amputation	1360.71
Fracture	574.87
Foreign Body	325.72
Strain, Sprain	226.77
Hemorrhage	179.56
Poisoning	110.03
Hematoma	99.58
Dermat Or Conj	95.36
Dental Injury	79.17
Ingested Object	79.17
Nerve Damage	79.17
Burns, Thermal	38.74
Burns, Chemical	16.18
Inter Organ Injury	16.18

body_part	n
Finger	212292.96
Hand	59287.40
Lower Arm	10986.55
Wrist	5512.13
Lower Leg	5156.99
Upper Leg	4931.13
Foot	4792.77
Knee	1666.74
Lower Trunk	1563.82
Face	1462.47
Toe	1228.27
Upper Arm	828.96
Ankle	750.35
Upper Trunk	644.14
Eyeball	555.77
Mouth	379.93
Neck	336.24
Head	309.10
Elbow	263.10
Shoulder	183.52
All Of Body	110.03
N.S./Unk	94.66
Ear	88.69
Pubic Region	82.66
Internal	79.17

location	n
Home	214092.90
Unknown	93006.86
Sports Or Recreation Place	2841.22
Other Public Property	2307.34
School	1067.87
Street Or Highway	254.34
Farm	16.99

Fig.18 Resulting App

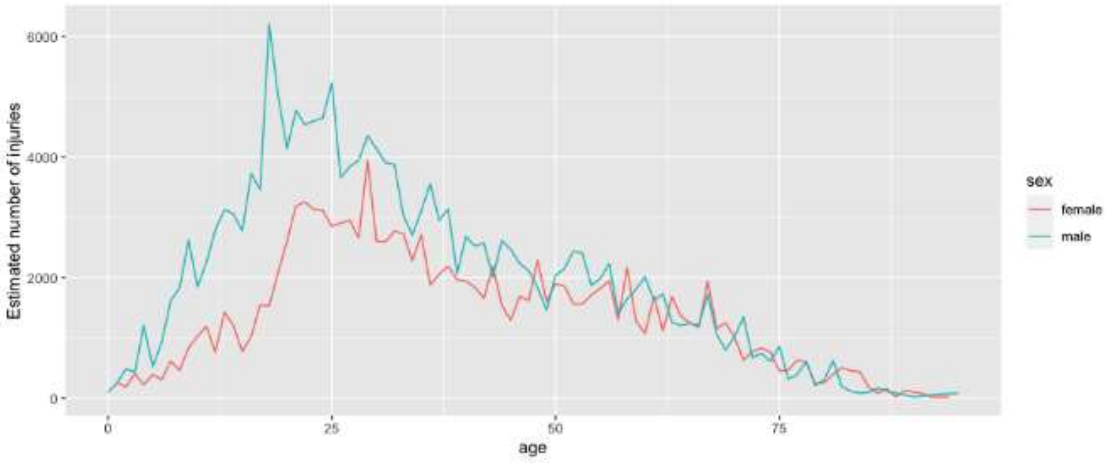

Fig.19 initial version of the NEISS exploration app

11. TERMINAL QUESTIONS

Short Type Questions:

1. What is R-Shiny?
2. What is reactivity?

Long Type questions:

1. What is Visual Analytics?
2. What is R-Shiny?Explain the two phases of R Shiny.

12. SELF ASSESSMENT ANSWERS

1. graphs, charts,maps
2. Visual Analytics
3. Web app
4. True
5. install.packages("shiny"))
6. Javascript
7. Create graphs and charts using the server function.
8. HTML

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13. TERMINAL QUESTIONS ANSWERS

Short Type Questions

1. What is R-Shiny?

   An R package called Shiny makes it possible to create interactive web applications that can run R code in the background. With Shiny, you can create dashboards, embed interactive charts in R Markdown papers, and host standalone applications on a website. Additionally, you can add HTML widgets, JavaScript actions, and CSS themes to your Shiny applications.

2. What is reactivity?

   Reactive programming is the method used by Shiny. This implies that everything that depends on something will immediately be updated when that item changes (such the slider being adjusted). The distPlot graph is the only component of the sample app that is dependent on the slider; as a result, if we change the slider, this will be redrew.

Long Type Questions

1. What is Visual Analytics?

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   Benefits:Share findings and monitor progress:Organize and share key performance indicators across an organisation by using interactive reports and dashboards.

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Few examples are

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Supply Chain:By displaying KPIs and enabling interactive exploration, big data visual analytics can assistsupply chain managers in quickly discovering relationships across complicated, divergent data sources.

Sales:The clear, organised presentation of sales data helps sales managers to boost revenue,

enhance forecasting, and spot important patterns.

Finance:A loan manager at a consumer bank can investigate how various geographic areas, products, and loan officers fare over time and determine which factors have the most influence on revenue and profits.

2. What is R-Shiny?Explain the two phases of R Shiny.

WHAT IS R SHINY?

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Installation:Installing the package is a prerequisite for using it in R programming.

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- Similar to how spreadsheets are automatically "active." Changes to the input will result in an automated update.
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The web application's code is located in the server function. A render...() function corresponds to each...Output() function in the user interface. The R code to create the object we wish to render is provided as the first argument to the render...() function. For instance, the code to create a histogram is located in the renderPlot() function of the example app. Since this must be a single expression, it is usually found in s.

The value of the slider as defined in the user interface will be contained in input$bins (which was called bins).

A SERVER FUNCTION

The web application's

code is located in the server function. A render...() function corresponds to each...Output() function in the user interface. The R code to create the object we wish to draw is provided as the first parameter to the render...() function.

For instance, the code to create a histogram is located in the renderPlot() method of the sample app. Since this must be a single expression, it is usually seen in s.

14. CONCEPT MAP

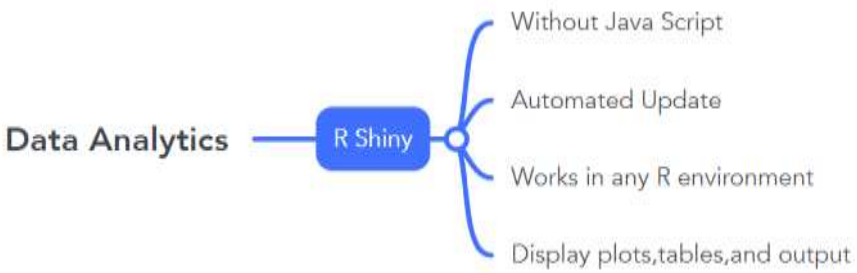

Fig 20.Mindmap of Rshiny

15. REFERENCES

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- https://www.analyticsvidhya.com/blog/2021/05/build-interactive-models-with-r-shiny/
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- https://www.qlik.com/us/data-visualization/visual-analytics